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NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester B.Tech (CBCS) Degree Examinations, December 2025**25MAT101 – MULTIVARIATE CALCULUS AND CURVE FITTING**

(For BTY, CIV, MEC)

Duration: 3 Hours]

[Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each Unit.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks****CO1:** Q1 to Q4; **CO2:** Q5 to Q8; **CO3:** Q9 to Q12; **CO4:** Q13 to Q16; **CO5:** Q17 to Q20.

- For the polar curve $r = f(\theta)$, the relation between θ and coordinates (x, y) is:
A) $\tan\theta = \frac{y}{x}$ B) $x = r\cos\theta$
C) $x = r\sin\theta$ D) $\tan\theta = \frac{x}{y}$
- Two polar curves intersect orthogonally if _____
A) $\tan\phi_1 \cdot \tan\phi_2 = \frac{\pi}{2}$ B) $\tan\phi_1 \cdot \tan\phi_2 = 1$
C) $\tan\phi_1 \cdot \tan\phi_2 = -1$ D) $\tan\phi_1 + \tan\phi_2 = 1$
- The radius of curvature of the curve $y = e^x$ at the point where it crosses the y-axis is
A) $2\sqrt{2}$ B) $\sqrt{2}$
C) 2 D) $\sqrt{2}/2$
- The Maclaurin's series expansion of the function $f(x) = e^x$ is:
A) $-1 + x - \frac{x^2}{2} + \frac{x^3}{6} - \dots$ B) $1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$
C) $1 - x - \frac{x^2}{2} - \frac{x^3}{6} + \dots$ D) $1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \dots$
- If $z = 3x^2y + 5y$, then $\frac{\partial^2 z}{\partial x \partial y}$ is _____
A) 6x B) 3xy
C) 5x D) 5xy
- If $f(x, y) = x^2y + 3y^2$, what is the value of f_x and f_y at (2,1)?
A) $f_x = 2, f_y = 10$ B) $f_x = 4, f_y = 9$
C) $f_x = 4, f_y = 7$ D) $f_x = 4, f_y = 10$
- If $u = x^2y + xy^2$, then according to Euler's theorem $xu_x + yu_y$ is _____
A) u B) 3u
C) 2u D) 4u
- If $r = f_{xx}, s = f_{xy}, t = f_{yy}$, then $f(x, y)$ has minimum at (a, b) if,
A) $rt - s^2 < 0$ & $r < 0$ at (a, b) B) $rt - s^2 > 0$ & $r < 0$ at (a, b)
C) $rt - s^2 < 0$ & $r > 0$ at (a, b) D) $rt - s^2 > 0$ & $r > 0$ at (a, b)

9. The value of $\int_0^1 \int_0^{\sqrt{x}} dy dx$ is
 A) $2/3$ B) $3/2$
 C) $1/3$ D) $1/2$
10. The value of $\int_0^3 \int_0^2 \int_0^1 8xyz dx dy dz$ is
 A) 3 B) 9
 C) 36 D) 4
11. If n is a positive integer then $\Gamma(n)$ equals
 A) $n-1$ B) $n!$
 C) $(n-1)!$ D) n
12. $\int_0^{\pi/2} \sin^7 x \cos^5 x dx$ is equal to
 A) $\beta(3,2)$ B) $\frac{1}{2}\beta(4,3)$
 C) $2\beta(4,3)$ D) $\beta(7,5)$
13. The velocity at time $t = 1$ of a particle moving along the curve $x = t, y = t$ and $z = -t$ is
 A) $\hat{i} + \hat{j} - \hat{k}$ B) $\hat{i} + \hat{j} + \hat{k}$
 C) 0 D) $\hat{i} + \hat{j}$
14. A unit tangent vector to the surface $x = t, y = e^t, z = -3t^2$ at $t = 0$ is
 A) $\hat{i} + \hat{j}$ B) $\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}\hat{j}$
 C) $\frac{1}{\sqrt{3}}\hat{i} + \frac{1}{\sqrt{3}}\hat{j}$ D) none of these
15. If $\phi(x, y, z) = c$ is a surface, then $\nabla \phi$ is
 A) parallel to the surface B) unit vector
 C) normal to the surface D) none of these
16. A vector field which has a vanishing curl is called as
 A) irrotational field B) scalar field
 C) rotational field D) Solenoidal field
17. One among the three normal equations for fitting of a parabola $y = a + bx + cx^2$ is ____
 A) $\sum xy = a \sum x + b \sum x^3 + c \sum x^4$ B) $\sum xy = a \sum x^2 + b \sum x^3 + a \sum x^4$
 C) $\sum xy = a \sum x + b \sum x^2 + c \sum x^3$ D) $\sum xy = a \sum x + b \sum x^3 + c \sum x^2$
18. If regression coefficient of x on y is 0.7 and that of y on x is 0.3 then correlation coefficient $r =$ ____
 A) 0.21 B) 0.4582
 C) 0.021 D) 0.04582
19. The standard deviation of x and y are 6 and 8 respectively. The correlation coefficient between them is 0.6. Then the value of regression coefficient b_{yx} is ____
 A) 0.45 B) 0.6
 C) 0.75 D) 0.8
20. In multiple linear regression, the dependent variable must be:
 A) Continuous B) Binary
 C) Categorical D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

	Unit - I	Marks	BT*	CO*	PO*
1. a). Show that $\tan \phi = r \frac{d\theta}{dr}$		6	L*2	1	1,2
b) Develop the pedal equation of the curve $\frac{2a}{r} = 1 - \cos \theta$		6	L3	1	1,2

c)	Make use of Maclaurin series expansion to expand $\sin x$ up to the terms containing 4th degree term.	4	L3	1	1,2
2. a)	Show that the curves $r = \frac{a}{1+\cos\theta}$ and $r = \frac{b}{1-\cos\theta}$ intersect each other orthogonally.	6	L2	1	1,2
b)	Obtain the radius of curvature at any point of the cardioid $r=a(1-\cos\theta)$	6	L3	1	1,2
c)	Make use of Taylor's series expansion to expand $\sin x$ in powers of $(x - \frac{\pi}{4})$ up to terms containing 4th degree term	4	L3	1	1,2

Unit – II

3. a)	If $u = f(2x - 3y, 3y - 4z, 4z - 2x)$ show that $\frac{1}{2} \frac{\partial u}{\partial x} + \frac{1}{3} \frac{\partial u}{\partial y} + \frac{1}{4} \frac{\partial u}{\partial z} = 0$	6	L2	2	1,2
b)	If $u = x + y + z$, $uv = y + z$ and $uvw = z$ Evaluate $\frac{\partial(x,y,z)}{\partial(u,v,w)}$	6	L3	2	1,2
c)	Determine the first and second order partial derivatives of $z = e^x \cos y$	4	L3	2	1,2
4. a)	If $z = f(x + ct) + \phi(x - ct)$ determine $\frac{\partial^2 z}{\partial t^2}, \frac{\partial^2 z}{\partial x^2}$	6	L3	2	1,2
b)	If the perimeter of a triangle is a constant, establish that, the area of this triangle is maximum when the triangle is equilateral	6	L3	2	1,2
c)	If u is a homogeneous function of degree n in x and y , then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$	4	L2	2	1,2

Unit – III

5. a)	Obtain the area bounded by the cardioid $r = a(1 + \cos\theta)$	6	L3	3	1,2
b)	Prove that $\beta(m, n) = \frac{\Gamma(m)\Gamma(n)}{\Gamma(m+n)}$	6	L2	3	1,2
c)	Evaluate $\int_0^1 \int_x^{\sqrt{x}} (x^2 + y^2) dy dx$	4	L3	3	1,2
6. a)	Evaluate $\int_0^\infty \int_0^x x e^{-x^2/y} dy dx$ by changing the order of integration.	6	L3	3	1,2
b)	Determine the volume common to the cylinders $x^2 + y^2 = a^2$ and $x^2 + z^2 = a^2$	6	L3	3	1,2
c)	Show that $\int_0^{\pi/2} \sqrt{\sin\theta} d\theta \int_0^{\pi/2} \frac{1}{\sqrt{\sin\theta}} d\theta = \pi$	4	L2	3	1,2

Unit – IV

7. a)	A particle moves along the curve $x = 2\sin 3t$, $y = 2\cos 3t$, $z = 8t$, where t is the time variable. Obtain the magnitude of the velocity and acceleration at any time $t > 0$.	6	L3	4	1,2
b)	Evaluate $\text{div} \vec{F}$ and $\text{curl} \vec{F}$ at the point $(1,2,3)$ where $\vec{F} = x^2 yz \hat{i} + xy^2 z \hat{j} + xyz^2 \hat{k}$	6	L3	4	1,2
c)	Prove that $\nabla \cdot (\nabla \times \vec{F}) = 0$ Where $\vec{F}$ is a vector function.	4	L2	4	1,2
8. a)	Obtain the angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at $(2, -1, 2)$.	6	L3	4	1,2
b)	A particle moves along the curve $x = 2t^2$, $y = t^2 - 4t$, $z = -t - 5$, where t is the time variable. Determine the components of its velocity and acceleration at time $t = 1$ in the direction given by $\hat{i} - 2\hat{j} + 2\hat{k}$.	6	L3	4	1,2

- c) Find the constants a, b, c so that the vector field

$$\vec{F} = (axy + z^3)\hat{i} + (bx^2 + z)\hat{j} + (bxz^2 + cy)\hat{k} \text{ is irrotational.}$$

4 L3 4 1,2

Unit – V

9. a) Compute the rank correlation coefficient for the following data

x	68	64	75	50	64	80	75	40	55	64
y	62	58	68	45	81	60	68	48	50	70

6 L3 5 1,2

- b) Obtain the lines of regression for the following data.

x	1	2	3	4	5	6	7	8	9
y	9	8	10	12	11	13	14	16	15

6 L3 5 1,2

- c) Fit a straight line $y = ax + b$ to the following data by the method of least squares:

x	12	15	21	25
y	50	70	100	120

4 L3 5 1,2

10. a) If θ is the acute angle between the two regression lines in the case of two variables x and y , show that $\tan\theta = \frac{1-r^2}{r} \frac{\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2}$. Discuss the significance of value of r .

6 L3 5 1,2

- b) Compute Y when $X_1 = 10$ and $X_2 = 6$ from the least squares regression equation of Y on X_1 and X_2 for the following data

Y	90	72	54	42	30	12
X_1	3	5	6	8	12	14
X_2	16	10	7	4	3	2

6 L3 5 1,2

- c) Ten competitors in a beauty contest are ranked by two judges in the following order. Compute the coefficient of rank correlation.

I	1	6	5	3	10	2	4	9	7	8
II	6	4	9	8	1	2	3	10	5	7

4 L3 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
